NNTI PROJECT

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Abstract

Language Identification from Audio is an important task for multilingual speech systems. In this project, we classify short speech segments into one of 22 Indian languages using a dataset which has only 5 speakers per language, which introduces strong speaker bias. First, we reproduce the baseline on fine-tuning the pre-trained MMS-300M speech transformer (Pratap et al., 2023), achieving a validation accuracy of 31% and then improve the baseline through hyperparameter tuning and extended training to 37.57%. Finally, our Task 2 model reaches 70.54%. To mitigate speaker bias, in Task 2, we implement a speaker-disjoint train-validation split to ensure that speakers in validation set are not seen during training. Additionally, we apply light waveform augmentations including gain variation and additive noise and introduce regularization techniques such as dropout and early stopping. The feature extractor is frozen to prevent overfitting on the small dataset. The results show improved robustness and provide insights into the challenges of distinguishing acoustically similar languages.

1 Introduction

Language Identification from Audio is an important task in speech processing. It allows systems such as voice assistants, call center automation, and multilingual speech applications to detect the language being spoken before further processing. However, language identification can be challenging when the available training data is limited or when languages have similar acoustic properties.

In this project, we classify speech samples into one of 22 Indian languages. India has a large number of languages, many of which share similar phonetic and acoustic characteristics. Because of these similarities, it can be difficult for a model to correctly distinguish between languages using only short audio recordings.

Another important challenge comes from the dataset used in this project. Each language contains recordings from only five speakers due to which a model may learn to recognize the speakers instead of learning the actual characteristics of the language. This problem is known as *speaker bias*, and it can reduce the ability of the model to generalize to new speakers.

To solve this task, we start with a baseline system based on fine-tuning the pre-trained MMS-300M speech transformer (Pratap et al., 2023). This model was trained on large multilingual speech data and provides a strong starting point for speech classification tasks.

We first reproduce the baseline system and improve it by tuning the training hyperparameters and training for a longer time. To reduce speaker bias, we create a speaker-disjoint train–validation split, which ensures that speakers in the validation set do not appear in the training set. We also apply simple audio augmentation methods such as gain variation and additive noise, and use regularization techniques including dropout and early stopping to improve generalization.

Finally, we analyze the performance of the model and study which languages are most often confused with each other. This helps us better understand the challenges of spoken language identification when only limited training data is available.

2 Dataset

We use the NNTI spoken language dataset containing recordings from 22 Indian languages. The dataset has a total of 8,689 audio samples. Most languages have around 400 recordings, with a few having slightly lower namely, Bengali (379), Odia (373), and Konkani (337).

Each language contains recordings from exactly five speakers. This small number of speakers introduces a strong risk of speaker bias. A model may learn to recognize the speakers instead of the language itself, leading to poor generalization to unseen speakers.

The audio segments are short speech recordings spoken by a single speaker. All recordings are resampled to 16 kHz and processed using the MMS-300M feature extractor (Pratap et al., 2023).

To address the speaker bias, we create a speaker-disjoint train–validation split. This ensures that speakers in the validation set are not present in the training set. This setup allows the model to focus on language-specific patterns rather than speaker characteristics.

3 Method

In this project, we use a pre-trained **MMS-300M** speech transformer (Pratap et al., 2023) for spoken language identification. The baseline model fine-tunes MMS-300M on the 22 Indian languages using standard supervised training. Input audio is resampled to 16 kHz and truncated or padded to a maximum duration of 7 seconds. Labels are mapped to integers for classification.

3.1 Baseline Model

The baseline follows standard fine-tuning:

- The feature extractor of MMS-300M converts raw audio into input features.
- Labels are assigned to each audio sample, and the data is fed into the model.
- A simple data collator pads batches to equal lengths.
- The model is trained for a few epochs with a small learning rate and batch size.
- Accuracy on the validation set is used to select the best model.

3.2 Improvements over Baseline

To improve the baseline, we implemented several strategies:

- **Hyperparameter tuning:** Increased training epochs, adjusted learning rate and batch size, and applied gradient accumulation to better utilize GPU memory.
- **Regularization:** Enabled dropout layers to prevent overfitting. Also, added early stopping which monitors validation accuracy to stop training when improvements plateau.
- **Freezing feature extractor:** The feature extractor was frozen to prevent overfitting on the small dataset.

3.3 Speaker Bias Mitigation (Task 2)

The dataset has only five speakers per language, introducing **strong speaker bias**. To address this:

- Implemented a **speaker-disjoint train-validation split**, ensuring speakers in the validation set do not appear in the training set.
- Applied **waveform augmentations** to training audio to improve generalization and reduce reliance on speaker-specific cues. These include small **gain adjustments** and **additive Gaussian noise** (torchaudio, 2024).
- No time-shifting was applied because it can distort language-specific acoustic cues.

3.4 Training Details

- **Model:** MMS-300M (Pratap et al., 2023)
- **Optimizer:** AdamW with weight decay
- **Learning rate:** 1e-5, **batch size:** 8, **gradient accumulation:** 4
- **Maximum epochs:** 30 (early stopping used)
- **Metric:** Top-1 accuracy

These improvements and bias mitigation strategies collectively aim to make the model focus on language patterns rather than speaker identity, resulting in more robust language classification.

4 Experiments

4.1 Experimental Setup

All experiments were implemented in Python 3.10 using the PyTorch deep learning framework (Paszke et al., 2019) and the HuggingFace Transformers library (Wolf et al., 2020). The dataset was loaded using the HuggingFace Datasets library (Lhoest et al., 2021).

Training was performed on the university computing cluster using an **NVIDIA Tesla T4 GPU with 16 GB of memory**. Mixed precision training (fp16) was enabled to reduce GPU memory usage and improve training speed.

The MMS-300M model (Pratap et al., 2023) was fine-tuned using the AdamW optimizer with weight decay (Loshchilov and Hutter, 2019). The learning rate was set to 1×10^{-5} with a batch size of 8 and gradient accumulation of 4 steps. The model was trained for up to 30 epochs with early stopping enabled to prevent overfitting.

4.2 Data Splitting

For the baseline system, we used the original dataset split provided with the dataset.

For the improved model, we created a speaker-disjoint train–validation split. This means that speakers appearing in the validation set were not present in the training set. This setup reduces speaker bias and helps the model learn language characteristics instead of speaker-specific patterns.

4.3 Evaluation Metric

Model performance was evaluated using classification accuracy. Accuracy measures the percentage of audio samples whose language is correctly predicted by the model. The best model checkpoint was selected based on validation accuracy.

5 Results

We evaluated the performance of the baseline and Task 2 model on their respective validation sets. Task 2 uses a speaker-disjoint validation set to prevent speaker bias, whereas the baseline uses the original validation split. Figures 1 and 2 show the training loss and validation accuracy curves for both models.

5.1 Training Loss

Figure 1 presents per-epoch training loss. The baseline model achieves lower training loss because it is trained and validated on overlapping speakers, allowing it to more easily memorize the training data. Task 2 shows higher training loss because waveform augmentations and the speaker-disjoint split introduce variability and make the model focus on language patterns rather than speaker-specific cues.

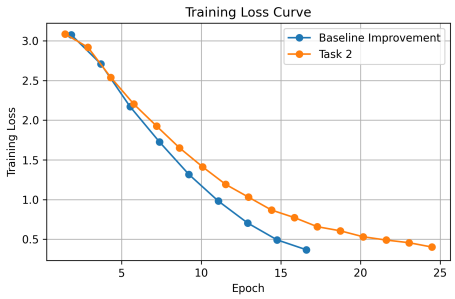


Figure 1: Training loss per epoch for Baseline and Task 2 models. Task 2 has higher training loss due to augmentations and speaker-disjoint splitting.

5.2 Validation Accuracy

Validation accuracy is higher for Task 2 (Figure 2), even though its training loss is larger. Specifically, the improved baseline reaches 37.57%, and the Task 2 model attains 70.54% accuracy. This is because Task 2 is evaluated on unseen speakers, providing a more realistic measure of generalization. The baseline's lower training loss does not indicate better generalization, since its validation set contains speakers seen during training.

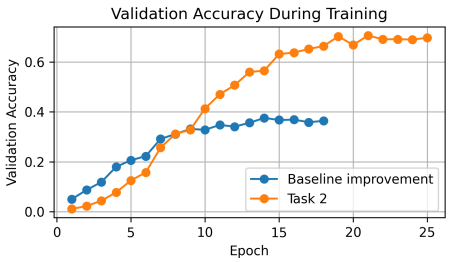


Figure 2: Validation accuracy per epoch. Task 2 achieves higher validation accuracy on unseen speakers, demonstrating better generalization.

6 Analysis

6.1 Source of Speaker Bias

The NNTI dataset contains only five speakers per language. This limited speaker variation introduces a strong speaker bias, where the baseline model can achieve low training loss by memorizing speakers rather than learning true language-specific patterns. As a result, generalization to unseen speakers is poor.

6.2 Effectiveness of Speaker-Disjoint Splitting and Augmentations

Task 2 mitigates speaker bias through a speaker-disjoint validation split and waveform augmentations such as gain variation and additive Gaussian noise (torchaudio, 2024). Although the training loss is higher (Figure 1), the validation accuracy improves (Figure 2), indicating that the model focuses on language characteristics rather than memorizing speakers.

6.3 Confusion Between Languages

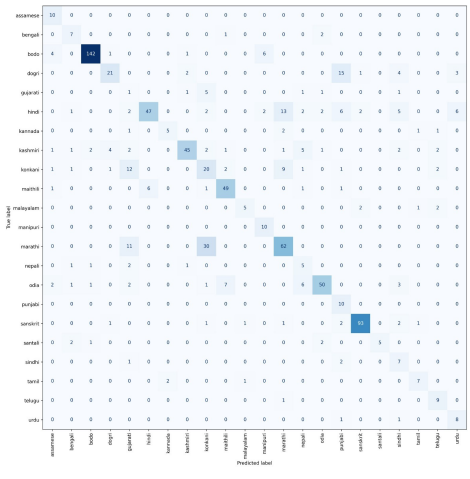


Figure 3: Confusion matrix for Task 2 on the speaker-disjoint validation set. Languages with similar phonetic characteristics are often confused.

Figure 3 shows the confusion matrix for our Task 2 model on the speaker-disjoint validation set. The highest off-diagonal counts occur between the following language pairs:

- **Dogri → Punjabi**: 15 Dogri samples misclassified as Punjabi.
- **Hindi → Marathi**: 13 Hindi samples misclassified as Marathi.
- **Konkani → Gujarati**: 12 Konkani samples misclassified as Gujarati.
- **Konkani → Marathi**: 9 Konkani samples misclassified as Marathi.
- **Marathi → Gujarati**: 11 Marathi samples misclassified as Gujarati.
- **Marathi → Konkani**: 30 Marathi samples misclassified as Konkani.
- **Odia → Maithli**: 7 Odia samples misclassified as Maithli.

These confusions reflect genuine acoustic similarities between the languages, as the validation set contains speakers unseen during training. The overall validation accuracy is 70.54%.

6.4 Representation Analysis

We visualized the last-layer embeddings using t-SNE. In the baseline model, the embeddings are fairly mixed in the speaker view, with only loose grouping in the language view, suggesting that the learned representation is still not very well organized. With the Task 1 improvement, the clusters become tighter and the language-wise structure starts to appear more clearly, although speaker overlap is still visible. In Task 2 with disjoint augmentation, the separation is the cleanest: clusters are more compact and align much better with language, indicating that the model is learning features that are more useful for distinguishing languages than speaker-specific traits. (Baseline model tsne analysis added in .zip)

6.5 Other Insights

Waveform augmentations increase variability during training, which explains the higher training loss. However, they improve robustness to unseen speakers. Early stopping prevented overfitting, particularly for languages with fewer samples. Overall, Task 2 provides a more realistic and generalizable model for spoken language identification.

7 Future Works

This project primarily utilized a dataset-centric approach to mitigate bias, In future, we plan to build

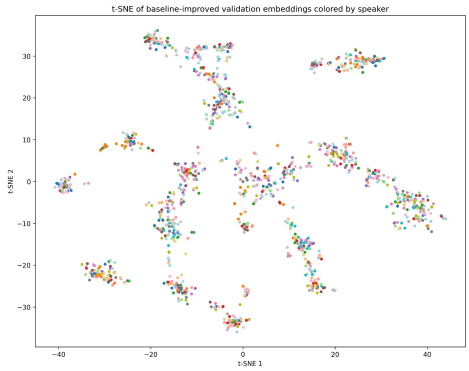


Figure 4: t-SNE of validation embeddings from the improved baseline model (Task 1), colored by speaker. Points cluster strongly by speaker.

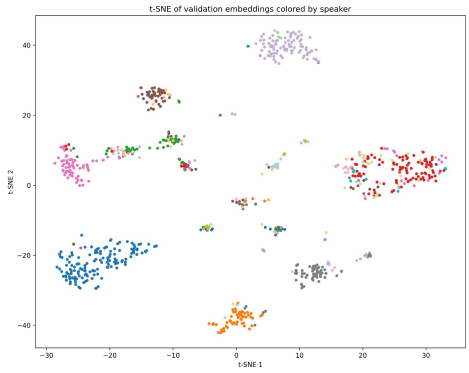


Figure 5: t-SNE of validation embeddings from the Task 2 model, colored by speaker. Points from different speakers are mixed, showing reduced speaker bias.

on our current data-focused approach by introducing a more balanced, hybrid strategy that combines data improvements with changes to the model architecture itself. Specifically, we are interested in a few key directions:

- Adversarial Speaker Normalization: By adding a branch that tries to identify the speaker and using a gradient reversal layer, we can encourage the model to ignore speaker-specific traits and focus on what really matters which is the linguistic content.
- Contrastive Learning: We want to explore loss functions like Triplet Loss or Supervised Contrastive Learning to help the model pull closer together different speakers of the same language in the embedding space, reinforcing language-level patterns over individual vocal characteristics.
- Temporal Modeling Enhancements: To better capture the flow and nuance of speech in Indian languages, we can consider integrating recurrent layers like LSTMs or more modern

Conformers on top of the MMS transformer, which could help the model track longer-range phonetic dependencies.

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